Jaymin Sheladia

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Education

University of Southern California — Los Angeles, CA
Master of Science in Computer Science — January 2026 – December 2027
- **Coursework:** Analysis of Algorithms, Agentic AI, Applied NLP, Web Technologies, Database Systems

Graphic Era Deemed to be University — Dehradun, India
B.Tech in Computer Science and Engineering — October 2021 – June 2025
- **Coursework:** Operating Systems, Computer Networks, Object-Oriented Programming, Software Engineering

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, HTML, CSS
Systems & Databases: PostgreSQL, MongoDB, MySQL, Redis, Database Internals, Query Optimization, Linux
Cloud & DevOps: AWS, Firebase, Docker, Kubernetes, Apache Kafka, Git, CI/CD, Microservices
Frameworks & Tools: React, Next.js, Node.js, FastAPI, Flask, Django, Express.js, REST APIs, GraphQL, GDB, GCC
AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, OpenCV, spaCy, OpenAI APIs, Claude APIs
AI Systems: Multi-Agent Orchestration, Workflow Automation, RAG, LangGraph, uAgents, pgvector
AI Dev Tools: GitHub Copilot, Cursor, Claude Code, ChatGPT, MCP, Claude/GPT APIs for agentic workflows

Work Experience

Cognizant Technology Solutions — Coimbatore, India
Programmer Analyst Trainee — May 2025 – September 2025
- Engineered a full-stack **travel planning and booking platform** using **Angular, Spring Boot, PostgreSQL, and Redis**, delivering **15+ RESTful APIs** with **JWT-based RBAC**, transactional persistence, and caching.
- Designed an **event-driven booking architecture** with **Apache Kafka**, Redis caching, and **Resilience4j circuit breakers**, reducing read latency by **30–40%** while preserving idempotent processing of concurrent booking requests.
- Integrated an **LLM-assisted recommendation engine** combining user preferences, budget, and trip duration to generate personalized itineraries; containerized services with **Docker** and achieved **80%+ test coverage**.

CODTECH IT Solutions — Remote
Machine Learning Intern — June 2024 – August 2024
- Developed an end-to-end **fraud detection system** for **200,000+** credit card transactions, establishing baselines for imbalanced data by benchmarking **XGBoost, Random Forest**, Logistic Regression, and Isolation Forest.
- Optimized model features and decision thresholds, achieving **90%+ fraud recall**; deployed the model through a **FastAPI** inference service with **real-time transaction risk scoring** and batch inference.
- Automated a **model monitoring and retraining pipeline** tracking **feature drift** and prediction quality, recording **under 100 ms** online scoring latency with Dockerized deployment and automated validation gates.

Projects

Haggle | GitHub | Python, uAgents, asyncio, Multi-Agent Orchestration, Sentry, Photon — October 2026
- Delivered **3 of 6 agents** for a **catering marketplace**, owning discovery and co-developing **test payment checks**, **idempotent bookings**, and **food rescue**; won **Fetch.ai runner-up honors** at **LA Hacks AI Hackathon 2026**.
- Implemented **parallel API calls** to Google Places and the agent registry with **asyncio**, radius filters, and **TTL caching**, retaining results if one source failed; added **Sentry tracing** and **Photon booking notifications**.

TechHire | GitHub | Python, FastAPI, LangGraph, React, pgvector, Redis — June 2026 – September 2026
- Built a **RAG pipeline** for **1,040 job postings** with **pgvector embeddings** and PostgreSQL full-text search, combining semantic and keyword retrieval and raising benchmark **top-10 hit rate from 68% to 92%**.
- Orchestrated **3 concurrent LLMs** using **LangGraph** for fault-tolerant summaries and **citation-checked answers**; reduced benchmark query latency **13.8x** (**19.1 ms to 1.38 ms**) through **Redis caching** for repeated searches.

Model Router | GitHub | Python, FastAPI, SQLite — July 2026 – August 2026
- Architected a **policy-aware FastAPI router** across **3 LLM tiers** with project rules, **transactional budget reservations**, and SQLite audit logging, enabling **automatic failover** to lower-tier models during service failures.
- Evaluated heuristic routing on **90 hand-labeled prompts** using real Groq calls, achieving **97.8% tier-selection accuracy (88/90)** and **36.4% lower benchmark inference cost** compared with always using the frontier model.

Research & Publications

Deep Learning Approach for Malicious URL Detection | Published Research Paper — 2025
- Compared **4 deep learning architectures (CNN, RNN, LSTM, Bi-LSTM)** for malicious URL detection using **character-level preprocessing** on the IEEE Dataport dataset, establishing accuracy and error-rate baselines.
- Achieved **99.25% accuracy with Bi-LSTM**, outperforming CNN (**86.5%**), LSTM (**63.75%**), and RNN (**53%**) baselines across classification and error-rate metrics.